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// Circular Queue Implementation Using Array for Parcel Handling System

#include<stdio.h>
#define size 5
void main(){
 int arr[size];
 int op;
 int front = -1, rear = -1;
 do{
  printf("1. Add\n2. Dispatch\n3. Display\n4. Exit\n");
  printf("Enter a option: ");
  scanf("%d", &op);
  switch(op){
     case 1:
     if(front == (rear + 1)%size){
      printf("Parcel queue is Full\n\n");
      break;          }
     else{
      if(front == -1){
     front++;
    }
     rear = (rear+1)%size;
     printf("Enter parcel no. to add: ");
     scanf("%d", &arr[rear]);
     printf("\n");          }
    break;
     case 2:
      if(front == -1){
       printf("Parcel queue is Empty\n\n");
       break;          }
      else{
       printf("Parcel %d is dispatched\n\n", arr[front]);
       if(front == rear){
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            front = -1;
            rear = -1;              }
          front = (front+1)%size;           }
        break;
       case 3:
        if(front == -1){
          printf("Parcel queue is empty\n\n");
          break;
        }else{
          int i = front;
          printf("Parcel no. left in queue are: ");
          do{
            printf("%d ", arr[i]);
            i = (i+1)%size;              }
         while(i != (rear+1)%size);
        printf("\n\n");           }
        break;
       case 4:
        printf("EXITED SUCCESSFULLY");
        break;       }    }
   while(op <= 3);
}
//Output
1. Add
2. Dispatch
3. Display
4. Exit
Enter a option: 1
Enter parcel no. to add: 1

1. Add
2. Dispatch
3. Display
4. Exit
Enter a option: 1
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Enter parcel no. to add: 2

1. Add
2. Dispatch
3. Display
4. Exit

Enter a option: 1

Enter parcel no. to add: 3

1. Add
2. Dispatch
3. Display
4. Exit

Enter a option: 1

Enter parcel no. to add: 4

1. Add
2. Dispatch
3. Display
4. Exit

Enter a option: 1

Enter parcel no. to add: 5

1. Add
2. Dispatch
3. Display
4. Exit

Enter a option: 1

Parcel queue is Full

1. Add
2. Dispatch
3. Display
4. Exit

Enter a option: 2

Parcel 1 is dispatched

1. Add

2. Dispatch

3. Display

4. Exit

Enter a option: 3

Parcel no. left in queue are: 2 3 4 5

1. Add

2. Dispatch

3. Display

4. Exit Enter a option: 4

EXITED SUCCESSFULLY